Flight Delay & Cancellation Analysis

SECTOR : AVIATION

NEWTON SCHOOL OF TECHNOLOGY

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SECTOR: AVIATION

STAKEHOLDER: AIRLINE OPERATIONS MANAGERS & REGULATORS

Context & Problem Statement

Sector Overview

The US domestic aviation sector handles 900M+ passengers annually across 3M flight operations. Despite sustained investment, on-time performance remains a persistent challenge.

Core Business Question

> *"Which airlines, routes, seasons and delay causes drive the worst on time performance for Airline Operations Managers, Aviation Regulators (FAA/DOT), and Airport Capacity Planners?"*

Objective

Enable airline operations managers to prioritize which delay causes, routes and carriers require targeted intervention to improve on time performance and reduce cancellations.

Business Impact:

- Reduce operational costs
- Improve customer satisfaction
- Optimize resource allocation
- US DOT estimates annual passenger delay costs exceed $30 billion

Data Engineering

Dataset Overview

Source: Kaggle Flight Delay and Cancellation Dataset (2019-2023)

Records: 300,000 (sampled from 3M)

Period: Jan 2019 - Dec 2023

Columns: 32 attributes

Format: CSV

Data Quality Issues

- **cancellation_code:** 97.4% missing; only present for cancelled flights
- **delay_due_* columns:** 82.2% missing; only populated when delay ⩾15 min

Major Cleaning Steps

- Removed nulls
- Handled outliers (468 rows ⩾500 min removed)
- Validated IATA codes
- Standardized dates
- Extracted temporal features
- Created delay categories

KPI Framework

On-Time Arrival Rate (%)

82.44% (flights ⩽15 min late)

Delay Rate (%)

17.56% (flights >15 min late)

Cancellation Rate (%)

2.63%

Median Arrival Delay

41.0 minutes (50% of flights arrived ⩽41 min late)

Controllable Delay Share (%)

73.6% (3 in 4 delay minutes preventable)

Why These Metrics Matter

- On-Time Arrival Rate is the FAA industry-standard DOT benchmark.
- Delay Rate shows operational performance.
- Cancellation Rate captures total service disruption.
- Median Arrival Delay indicates severity assessment.
- Controllable Delay Share shows intervention priority and responsibility.

Key EDA Insights

1. **Operational Scale:** 299K+ flights, **17.56% delay rate**, **2.63% cancellation rate**.
2. **Delay Distribution: 82.44% on-time**, **14.29% moderate** (15-60 min), **3.27% severe** (60+ min).
3. **Carrier Variance: 12.6%** (Endeavor) to **22.4%** (JetBlue) = **2x gap**.
4. **Seasonality: Jun-Aug peak** with **5.86 min** additional delay vs. rest of year.
5. **Day-of-Week: Friday 19.2%** delay rate vs. **Tuesday 15.0%** (**3.2x higher**).
6. **Route Congestion:** High-traffic hubs (**EWR, SFO, BOS, MCO**) show elevated delays.
7. **Root Cause: 73.6%** of delays from **Carrier (42.1%)** + **Late Aircraft (31.5%)**.
8. **Cancellation Drivers: Weather 36.2%, Security 31.4%, Carrier 24.9%.**

Key Takeaway: Flight disruptions are driven by interconnected factors: carrier efficiency, route congestion, seasonal demand, and weather — with significant improvement potential through operational optimization.

Advanced Analysis

Statistical Methods Applied

- **Pearson Correlation:** Measured relationships between departure delay, arrival delay, and distance. Found **r = 0.9368** (departure to arrival) and **r ≈ -0.0005** (distance effect).
- **Hypothesis Testing (t-test & ANOVA):** Validated summer seasonality effect (**t = 36.09**, $p < 0.001$), airline performance gap (**t = 22.44**, $p < 0.001$), and day-of-week effect (**F = 55.94**, $p < 0.001$).
- **OLS Regression:** Predicted arrival delay from operational variables with **R^2 = 0.9342**, explaining **93.4%** of variance.

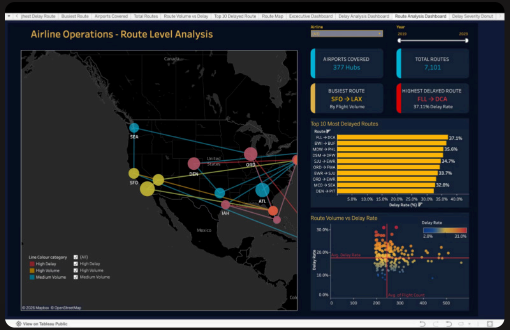
Key Findings

- **Delay Propagation Confirmed:** Departure delay and arrival delay show **very strong correlation (r = 0.93)** — late departures almost always cause late arrivals.
- **Distance Not a Driver:** Flight distance has near-zero relationship with delays, proving delays are mainly ground-operation issues.
- **Seasonality is Real:** Summer months (Jun–Aug) have significantly higher average delays than non-summer months ($p < 0.05$).
- **Carrier Performance Gap:** Best vs worst airline delay performance differs significantly; worst carrier averages **13.89 mins** more delay.
- **Weekday Effect:** Friday flights carry higher delay risk than Tuesday flights (*ANOVA* $p < 0.05$).
- **High Predictability:** Regression model explains **93.4% of arrival delay variance (R^2 = 0.934)**.
- **Controllable Impact: 73.6% of delay minutes** come from Carrier + Late Aircraft causes, making delays largely operationally reducible.

> Strategic Takeaway*:* **Flight delays are highly predictable and largely controllable — improving departure operations, staffing, scheduling, and aircraft turnaround can deliver the biggest performance gains.**

Dashboard Overview

Tableau Dashboard - Three Views



Executive View

- KPI cards
- 5-year trends
- Top/bottom performers
- Seasonal patterns

Operational View

- Airline/date/airport filters
- Route analysis
- Cancellation breakdown
- Day-of-week patterns

Route & Airport Network Analysis

- US route network map colored by delay and sized by volume
- Top 10 delayed routes with 30%+ delay rate
- Volume vs delay correlation: higher volume = better performance

Dashboard Access: Tableau Public URL - **https://public.tableau.com/app/profile/preetish.ubhrani/viz/Flight_Delay_Analysis_Final_Dashboard/HomePage**

INSIGHTS

OPERATIONAL DELAY ANALYSIS

Top Insights

01

Delays Are Operationally Determined, Not Structural

Distance has near-zero correlation with arrival delay ($r \approx -0.0005$)

02

Departure Efficiency Compounds into Arrival Recovery

Near-perfect correlation ($r = 0.937$) means every minute recovered at gate translates to 0.94-minute arrival improvement

03

Summer Months Are Predictably Worse

Jun-Aug carries 5.86 minutes additional delay vs rest of year ($p < 0.001$)

04

Friday Is the Riskiest Day

Friday delays 3.2x higher than Tuesday (4.82 min vs 0.79 min)

05

13.89-Minute Performance Gap Between Best and Worst Carriers

Endeavor Air (-2.65 min) vs JetBlue (11.24 min)

06

Controllable Causes Drive 73.6% of Delay Minutes

Carrier (42.1%) + Late Aircraft (31.5%) together account for 73.6%

07

NAS Delays Have Highest Per-Minute Impact

Regression coefficient 0.90 means each NAS minute causes 0.90 arrival delay minutes

08

Regression Model Explains 93.4% of Delay Variance

R^2 of 0.9342 confirms delays are highly predictable from operational inputs

Recommendations

Implement Airline-Specific Delay Reduction Programs

Establish peer benchmarking groups. Bottom-quartile carriers (JetBlue, Frontier, Spirit) compare practices against top-quartile (Endeavor, Hawaiian, Southwest). Focus on gate turnaround times, crew scheduling algorithms, maintenance positioning. Expected impact: JetBlue reducing from 11.24 to 6.24 min avg delay = 2.5M+ passenger-hours recovered.

Seasonalise Staffing and Buffer Schedules

Increase ground staffing by 8-12% during Jun-Aug. Add 4-6% buffer to block times during summer. Front-load maintenance during Apr-May. Expected impact: Recover 1-2 minutes of average delay during peak months.

Prioritise Departure Gate Performance Management

Deploy real-time gate utilisation dashboards. Implement predictive pushback algorithms. Set target: 95% of flights depart within 5 min of scheduled push-back. Expected impact: 2-3 min reduction in mean departure delay.

Target Late Aircraft Cascade Reduction

Implement minimum ground time buffers of 25-30 minutes for back-to-back routes. Prioritise buffer insertion on high-frequency routes (SFO→LAX, ORD→LGA). Expected impact: 20% reduction in late aircraft delays would recover ~256,000 delay minutes annually.

Impact & Feasibility

Impact Estimation

Recommendation	Impact Type	Estimated Impact
Airline Benchmarking Program	Cost Savings	\$18–25M per major carrier annually through reduced crew overtime, fuel burn, and maintenance repositioning.
Seasonal Staffing Adjustment	Efficiency Gain	1–2 min delay reduction per flight during Jun–Aug peak season; improved crew utilisation rates.
Gate Performance Management	Cost + Efficiency	\$8–12M per carrier annually; 2–3 min reduction in mean departure delay.
Day-of-Week Scheduling	Revenue + Efficiency	Demand smoothing across weekdays; marginal revenue uplift from dynamic pricing.
Controllable Delay Framework	Risk Reduction	2–3 min system-wide delay reduction; improved regulatory compliance positioning.

Limitations

Sample Representativeness

The analysis uses a stratified sample of 300,000 flights drawn from approximately 3 million flights across 5 years. Rare events such as severe weather and security incidents may be underrepresented relative to their true frequency.

External Factors Not Captured

The dataset does not include fuel prices, crew contract changes, equipment availability changes over time, or pandemic-related operational restructuring. The 2020–2021 period also introduces structural breaks in traffic and delay patterns that are not modelled separately.

Causality vs Correlation

The regression model quantifies associations, not causal effects. Reverse causality is possible — later scheduled flights may carry slack buffers that inflate observed delay correlations rather than cause them.

Airline-Specific Context

Performance gaps may reflect route mix differences, such as carriers focusing on delay-prone hub routes; network effects, where hub-and-spoke carriers are more susceptible to cascading delays; and aircraft age or type differences.

Self-Reported Data

Cancellation codes and delay cause attribution are self-reported by carriers to BTS. Classification bias may exist, particularly between Carrier (A) and NAS (C) causes.

NEXT STEPS

Future Extensions

1. **Predictive Modeling:** Build ML models (Random Forest/XGBoost) for 7–14 day delay forecasting.
2. **Data Integration:** Connect real-time weather APIs, maintenance records, and crew schedules.
3. **Causal Analysis:** Apply causal inference techniques to isolate true drivers.
4. **Automation:** Deploy automated alerts for anomaly detection.

Closing Summary

- On-time performance challenge is operational, not structural or weather-driven.
- 73.6% of delay minutes are within airline control.
- Departure efficiency near-perfectly predicts arrival performance (r = 0.937).
- Statistically proven 13.89-minute performance gap separates best and worst carriers.
- Recommended actions are specific, measurable, and operationally grounded.
- Top performers are already implementing these solutions.
- The data is clear: aviation can do better.
- Analytical foundation has been laid.
- Next step is operational commitment.

Thank you